



NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
First Semester M.C.A. (CBCS) Degree Examinations

January 2024

22MCA102 – ADVANCED DATABASE SYSTEMS

Duration: 3 Hours

Max. Marks: 100

Note: Answer Five full questions choosing One full question from each Unit

		Unit – I	Marks	BT*	CO*	PO*
1.	a)	Discuss the main characteristics of the database approach and how it differs from traditional file system.	10	L*2	1	1
	b)	Explain structural constraints of relationships in ER Model.	10	L2	1	1
2.	a)	Explain entity types with example.	10	L2	1	1
	b)	Explain the types of attributes in ER Model. Give examples for each.	06	L2	1	1
	c)	List all notations used in ER Diagram.	04	L1	1	1
		Unit – II				
	a)	Give the syntax and examples of the following SQL commands: i) DROP ii) ROLLBACK iii) UPDATE iv) ALTER	08	L2	2	1
	b)	Discuss views in SQL. Give an example.	04	L2	2	1
	c)	Explain aggregate functions in SQL with suitable examples.	08	L2	2	1
4.	a)	What is GROUPBY and HAVING clause in SQL? Explain with an example.	10	L2	2	1
	b)	Consider the following tables: Works(Pname, Cname, Salary) Lives(Pname, Street, City) Located_in(Cname, City) Description: Pname: Person name Cname: Company Name Write the SQL queries for the following: i) List the names of people who work for the company 'Wipro' along with cities they live in. ii) Find the names of persons who do not work for Infosys iii) Find people whose salaries are more than that of all of the 'Oracle' employees. iv) Find the persons who work and live in the same city.	10	L3	2	2
		Unit – III				
5.	a)	Differentiate between implicit and explicit cursors. Explain explicit cursors with their attributes.	12	L2	3	1
	b)	Explain PL/ SQL block structure with example.	08	L2	3	1
6.	a)	Explain stored procedures with example.	10	L2	3	1
	b)	Explain trigger syntax and trigger events.	10	L2	3	1
		Unit – IV				
7.	a)	Explain the desired properties of transactions.	10	L2	4	1
	b)	What are normal forms? Explain the first three normal forms.	10	L2	4	1

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a)	Write an algorithm to check whether the decomposition is Lossless or not. Justify with an example.	10	L2	4	1
b)	Write a note on functional dependency.	05	L2	4	1
c)	List the different informal design guidelines for relational schema.	05	L2	4	1

Unit – V

9.	a)	Explain types of NoSQL databases with their key features.	10	L2	5	1
	b)	Discuss the features of MongoDB. Explain update, remove and find methods with respect to MongoDB.	10	L2	5	1
10.	a)	Differentiate between SQL, NoSQL and NewSQL.	10	L2	5	1
	b)	What are the advantages of MongoDB? Explain insert and save methods with respect to MongoDB.	10	L2	5	1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
